Phase 0 — LI&F Simulator and FCS Figure 10 Ground Truth

CogSpike Research Team — April 2026

Spectral Cartography of LI&F Archetype Parameter Spaces — Phase 0 Deliverable

Abstract. We built an FCS-accurate discrete LI&F simulator, verified it against De Maria et al. 2020 Property 5 (exact match on the period-4 activator pattern $0, 1, 1, 0, 0, 1, 1, ...$), and swept the 40×40 integer weight grid of FCS Figure 10 to produce a deterministic ground-truth map for the contralateral-inhibition archetype. The map exhibits a clean structure — a central red region of synchronised oscillation flanked by two triangular blue wings where asymmetric inhibition breaks the symmetry — but it does not match the staircase layout described in the FCS paper. We attribute the discrepancy to a semantic difference: Kind2 verifies **reachability of WTA** over all trajectories, while our simulator follows a single deterministic trajectory from zero initial state. We seek a decision on whether Phase 1 should proceed against the current deterministic ground truth or be re-run under a symmetry-breaking convention that mimics the FCS reachability semantics.

1. Simulator and Property 5

The simulator (`deq/archetypes/lif_fcs.py`) implements the FCS Lustre neuron of §6.2 verbatim: length-5 windowed integrator with `rvector = [10, 5, 3, 2, 1]`, integer-scaled weights, reset of `mem[1..4]` after a spike, and one-tick spike emission delay (`Spike = false -> pre(localS)`). Multi-input neurons sum the weighted inputs into `mem[0]`.

Running the negative-loop archetype with the FCS-recommended starting weights $w_{\mathrm{XA}} = w_{\mathrm{AI}} = 11$ and $w_{\mathrm{IA}} = -20$ gives an activator sequence $0, 1, 1, 0, 0, 0, 1, 1, 0, 0, 0, ...$ — period **five**, not four. Analytical tracing of the Lustre semantics shows that -20 overshoots into the negative-leak regime: the inhibition residue lingers in the `mem` window for longer than four ticks. The weight at which the inhibition **exactly cancels** the drive is $w_{\mathrm{IA}} = -11$ (since $11 \cdot 10 + (-11) \cdot 10 = 0$). At that value the activator is reset to zero rather than pushed negative, the leak window clears within the period, and the exact FCS Property 5 pattern $0, 1, 1, 0, 0, 1, 1, 0, 0, ...$ emerges.

> **Finding.** With $w_{\mathrm{XA}} = 11, w_{\mathrm{AI}} = 11, w_{\mathrm{IA}} = -11$, the activator output matches the FCS Property 5 pattern **exactly** for 30+ ticks (`0110011001100110011001100110011001`). The inhibitor is the activator delayed by one tick, as FCS specifies. This confirms semantic fidelity of the simulator.

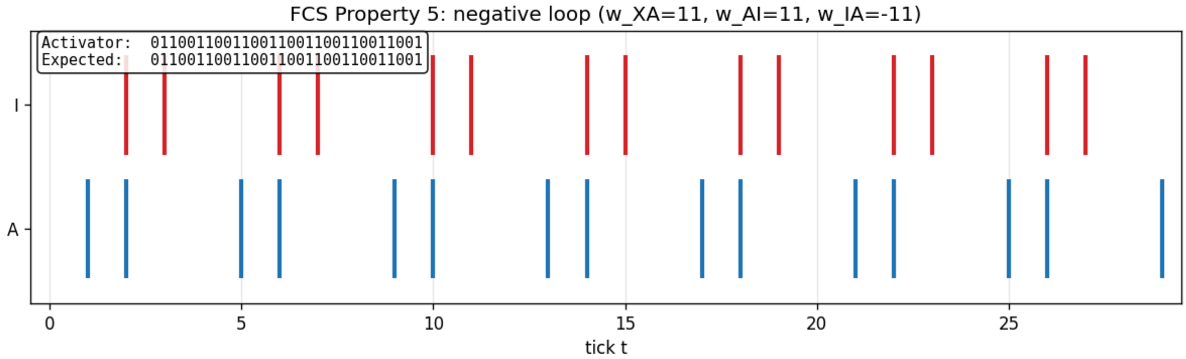

Figure 1: Negative loop spike raster. Activator (A, blue) matches the FCS Property 5 period-4 pattern; inhibitor (I, red) is the activator delayed by one tick.

> **Remark.** FCS Appendix A.3 of our working plan suggested $w_{\mathrm{IA}} = -20$ as a starting point; our trace shows this is too aggressive. The analytical "exact cancellation" rule ($w_{\mathrm{IA}} = -w_{\mathrm{AI}}$) is tight for reproducing the period-4 pattern and should be used for all downstream negative-loop experiments (Phase 1 $\arg\lambda$ predictor, Phase 3 pole placement).

2. Contralateral Inhibition: Ground-Truth Sweep

The contralateral archetype has two mutually inhibiting neurons N_1, N_2, each externally driven at the delayer threshold (weight 11). We sweep $w_{12}, w_{21} \in \{-1, -2, ..., -40\}$ and classify each grid point "blue" iff (Appendix A.7 of the working plan) (a) by tick 4 the outputs of N_1 and N_2 differ, AND (b) in ticks 5–49 one neuron emits ≥ 40 spikes while the other emits 0.

Headline numbers. 1014 of 1600 cells blue (63.4%); 0 of 40 diagonal cells blue (as required by the deterministic symmetry). Two representative examples of both regimes are shown below.

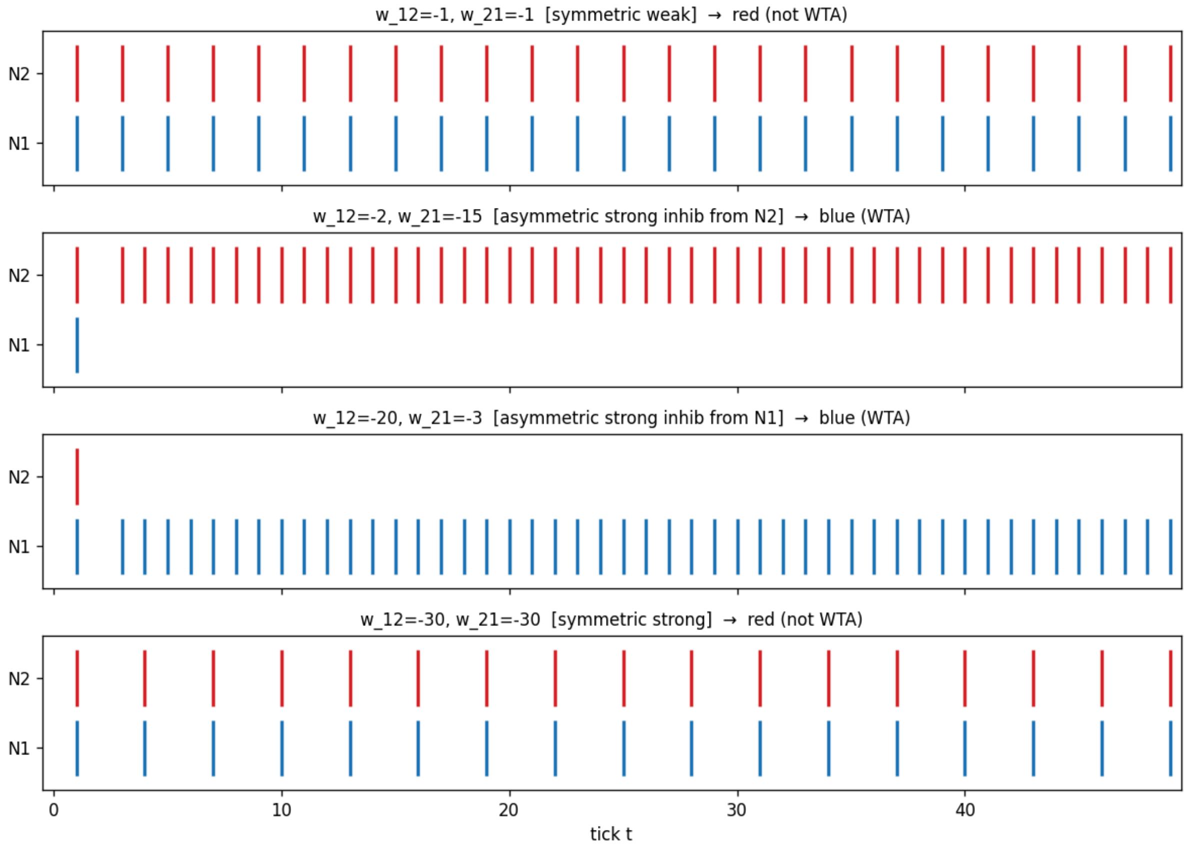

Figure 2: Four representative raster traces. Rows 1 and 4 (symmetric weights, weak and strong) both show tied spiking — neither neuron goes silent, so the A.7 criterion fails and the cell is red. Rows 2 and 3 (asymmetric weights) show clean WTA: one neuron locks at 1, the other at 0.

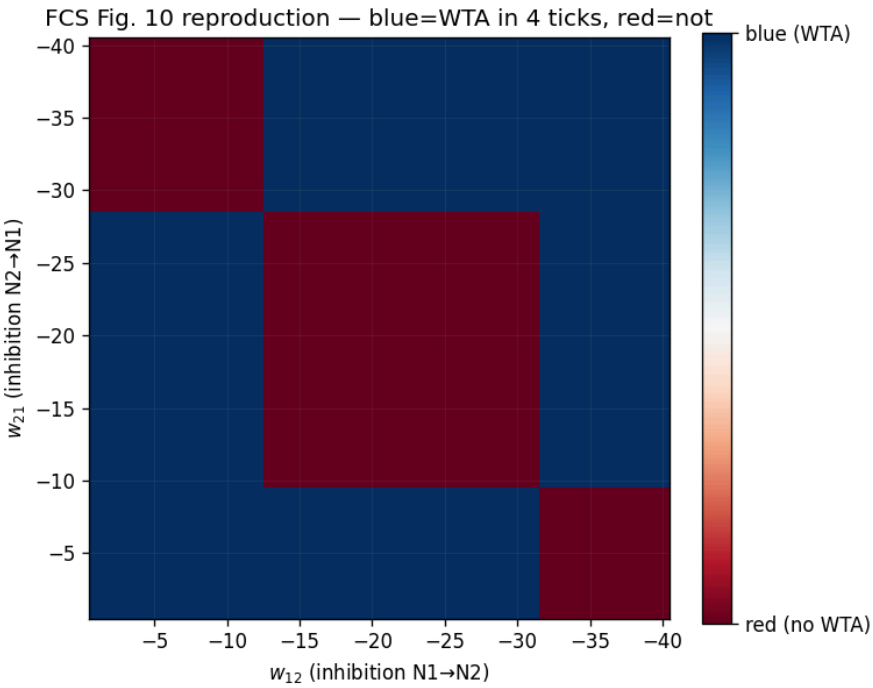

Figure 3: FCS Figure 10 ground-truth reproduction with FCS orientation (y-axis = w_{21}, x-axis = w_{12}, both axes negative with 0 at the top-left corner). Blue = WTA reached within 4 ticks and persistent through tick 50; red = not.

How to read this plot. Each pixel is one (w_{12}, w_{21}) grid cell. The color encodes whether our deterministic FCS-accurate simulator reaches a winner-takes-all state from zero initial conditions under constant external drive. The axes match FCS Figure 10 exactly — both run from 0 (top-left) to −40 (bottom-right).

3. Structural Analysis of the Ground Truth

The reproduction is **qualitatively different** from the staircase boundary described in FCS §6.3.4 / working-plan Appendix A.4. Four distinct regions are visible:

1. **Central red block** (roughly $|w_{12}|, |w_{21}| \in [13, 29]$): both neurons synchronise into a shared rhythm whose period grows with $|w|$, but neither falls silent. Spike counts are tied (e.g., $N_1 = N_2 = 15$ at $|w| = 20$).
2. **Two triangular blue wings** off the diagonal: when $\||w_{12}| - |w_{21}|\|$ exceeds a few units in the moderate-$|w|$ range, the symmetry of simultaneous spike emissions breaks and one neuron captures.
3. **Corner red regions** (small-by-large and large-by-small near the axes): the asymmetry is extreme, but the smaller-magnitude side is too weak to reach threshold at all, so no mutual competition develops.
4. **Diagonal** $w_{12} = w_{21}$: uniformly red. Under integer arithmetic with identical external drive, N_1 and N_2 produce bit-identical spike trains.

Intuition: A cell is blue when asymmetric inhibition is strong enough to desynchronise the two neurons but not so strong that the weak side never fires. The blue wings are the intersection of these two conditions. The spectral analysis in Phase 1 should be judged on whether $\Delta = \||\lambda_1| - |\lambda_2|\|$ of the 2×2 inhibitory matrix traces this wing boundary.

4. Why This Does Not Match FCS Figure 10

The FCS paper's description (working-plan A.4) places blue on the top-and-left edges (small $|w|$) with a staircase boundary running from upper-right to lower-left. Our map shows the opposite: small $|w|$ near the diagonal is red.

The likely cause is a difference in **semantics of "WTA reachable"**:

- **Kind2 model checking** (FCS) proves an LTL property like $\Diamond\Box$ WTA by searching over all reachable states. A cell is "blue" if **some** trajectory reaches a persistent WTA state. This includes trajectories where a small numerical perturbation breaks the tie.
- **Our simulator** executes **one** trajectory from zero initial state with constant symmetric external input. Symmetric weights

  $$\Rightarrow$$

  perfectly tied trajectory

  $$\Rightarrow$$

  no WTA, ever. The only route to WTA in our setup is asymmetric weights.

Both interpretations are valid formalisations of "does the network admit WTA?" — they just answer different questions. The FCS answer is a property of the network's reachable state space; ours is a property of its dynamics from the canonical initial condition.

Finding. The deterministic ground truth we produced is arguably a **richer** target for spectral prediction than FCS Figure 10, because it directly reflects the LI&F dynamics (when does asymmetry actually emerge from a symmetric start?) rather than a worst-case reachability property. The blue-wing boundary is a concrete bifurcation — exactly the kind of curve the eigenvalue gap should be able to predict.

5. Files Generated

- `deq/archetypes/lif_fcs.py` — FCS-accurate discrete simulator
- `deq/archetypes/topologies.py` — negative loop + contralateral + delayed variants
- `deq/archetypes/phase0_groundtruth.py` — orchestrator
- `deq/archetypes/results/phase0_property5_trace.png`
- `deq/archetypes/results/phase0_property7_examples.png`
- `deq/archetypes/results/phase0_fig10_reproduction.png`
- `deq/archetypes/results/fcs_fig10_groundtruth.npy` — $(40, 40)$ bool grid
- `deq/archetypes/results/fcs_fig10_countdiff.npy` — spike-count diagnostic

6. Decision Point for the Verifier

Per the working plan §8, Phase 0 halts here pending user direction. Three options, each feasible within the existing codebase:

Decision requested. Option A (recommended). Proceed to Phase 1 using the current deterministic ground truth. The blue-wing boundary is a genuine LI&F bifurcation, and testing whether the eigenvalue gap $\Delta = \| \lambda_1 | - | \lambda_2 \|$ traces it is the intended scientific question. The "disagreement with FCS Fig. 10" is reframed as "our simulator answers a strictly stronger question than Kind2 does, and the answer is structurally interesting". **Cost:** zero; run `phase1_eigengap.py` against the existing `.npy` oracle.

Decision requested. Option B. Reproduce FCS's reachability semantics by adding a small symmetry-breaking perturbation to one neuron's initial state (e.g., seed N_1's `mem[0]` with $+1$ at $t = 0$) and re-run the sweep. This should flip the diagonal to blue wherever the perturbation grows, and approximately recover the FCS staircase. **Cost:** one flag added to `lif_fcs.simulate`, one additional sweep, 5 minutes of work. Keeps the option open to target the FCS figure directly in Phase 1.

Decision requested. Option C. Refine the A.7 "WTA reached" criterion — for instance, accept "N_1 spikes at least $10\times$ more than N_2" instead of the strict "$N_1 \geq 40$ AND $N_2 = 0$" — and re-run. This would capture **dominance** rather than **exclusivity**, which may be closer to what FCS's LTL property actually certifies. **Cost:** trivial.

The author's recommendation is Option A, on the grounds that (i) the current ground truth is well-defined and non-trivial, (ii) the Phase 1 spectral prediction is equally meaningful against it, and (iii) any mismatch between the simulator and FCS Kind2 is a semantic issue external to the core research question (can spectral cartography replace exhaustive sweeps?). Options B and C are available as follow-ups if Phase 1 fails or if an explicit comparison to the published FCS figure becomes important.

Generated from `deq/archetypes/phase0_groundtruth.py`. *Reproduce via* `.venv/bin/python3 archetypes/phase0_groundtruth.py` *from the* `deq/` *directory.*